

NLP and use of AI in generating human-like text response

*Report submitted to Meharry Medical College
for the award of the degree*

of

Masters of Science

by

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Certificate of Capstone Project Completion

This is to certify that the report entitled **NLP and use of AI in generating human-like text**, submitted by **Isaac Adeoye Adebayo** to Meharry Medical College, is a record of bona fide research and practical work carried out under our supervision. The work presented in this report meets the academic and professional standards required for the successful completion of the Capstone Project as part of the Master of Data Science program at the Institute. In recognition of their dedication and scholarly contribution, this report is hereby deemed worthy of consideration for fulfillment of the degree requirements.

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- *To my parents* -

-I did it without the meds-

Declaration

I certify that

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- b. the work has not been submitted to any other institute for any other degree or diploma;
- c. I have followed the guidelines provided by the Institute in preparing the thesis;
- d. I have conformed to ethical norms and guidelines while writing the thesis;
- e. whenever I have used materials (data, models, figures and text) from other sources, I have given due credit to them by citing them in the text of the thesis, and giving their details in the references, and taken permission from the copyright owners of the sources, whenever necessary.

Isaac Adebayo

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I am grateful to my supervisor Dr Subash Neupane for guiding me throughout the implementation of my code and his constructive feedback to improve my understanding.

Executive Summary

The use of genAI generates revenue for companies in all sectors by cutting time, saving cost and improving efficiency. AI tools were around for decades, some common examples are Alexa and Siri which are voice based chat-bot that engage with users daily. Recently, the use of multi-modal chat-bots has increased drastically among all age demographic. Almost all common search engine in a web browser has an inbuilt AI scouring the web for information and making concise summary. However the use of AI come at a cost as technology involves. Advance models hallucinate more. For instance: OpenAI's newer reasoning models (o3, o4-mini) hallucinated 33% to 48% of the time, more than double the rate of their older, simpler models.

The goal of my project is to create a LLM based text chat-bot. The data used to train is derived from GitHub and is a medical dataset constituting of different clinicians' assessment, the goal is to train via LLM and retrieve data using RAG and incorporate this on a python user interface named Gradio design to test machine learning app (which will be hosted on hugging face). My chat-bot will serve as a clinical assistant that can retrieve information when prompted by a health care professional with appropriate access. The design of my text-based chat-bot shows that RAG can affect hallucination based on the type of data (structured vs unstructured data) and the frequency of word during embedding because of semantic closeness and temperature. My chat-bot shows RAG is far superior analyzing simple text file rather than data in a database

Keywords: RAG, Hallucination, LLM, Chat-bot.

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List of Abbreviations

LLM	Large Language Model
RAG	Retrieval augment Generation
API	Application Programming Interface
UI	User Interface
EDA	Exploratory data Analysis
GPT	Generated Pre-Trained Transformer
FAISS	Facebook AI Similarity Search

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Project Narrative

1 Introduction

The goal of my project is to build a LLM based chat-bot. My dataset is derived from GitHub and constitutes different clinicians' assessment, the goal is to train via LLM and retrieve data using RAG and incorporate this on a python UI GRADIO (which is hosted on hugging face). To achieve this, I will follow the following framework in creating the LLM based chatbot

- Perform Exploratory data analysis
- Choose an appropriate API model
- Chunking using recursive character text splitter
- Vector Embedding
- Storing vectors via FAISS
- User prompt
- Running Evaluation Metrics
- Hosting on Hugging Face using Gradio
- Testing by running a DEMO

1.1 Background

The project is a culmination that shows the foundational knowledge of NLP throughout the master's program. The project aim is to implement a modern approach in the development of a LLM based text chat-bot using various API and AI framework and tool. The focus will be on analyzing and implementing Retrieval Augmented Generation. This project creates an LLM based chat-bot and hosts it using a popular web interface designed for machine learning apps (Gradio), further analysis includes running multiple evaluation metrics to determine the precision, recall, and F1- score of the bot generated response to the user prompt and other metrics used to detect confidence and accuracy (coherence) in bot response. Assess common issues that leads to hallucination response, recommendation, and discuss limitations of modern AI framework that improves LLM accuracy. This project does not create an algorithm or function for retrieval from scratch but uses existing AI framework. It excludes any deep learning neural network model used in LLM training.

2 Objectives

The goal outcome for my simple chat-bot is to investigate the application of RAG in creating a chat-bot and analyze the role RAG plays in leading to hallucination based on the type of data it retrieves information from (structured vs unstructured data). The following objectives will be discussed :

- Create a crude LLM text based chatbot that is hosted on hugging face and test its functionality.
- Run evaluation metrics to determine precision, accuracy and F1 score of AI output using BERT, ROUGE AND BLEU, confidence threshold percentage, and fallback rate analysis to prevent model from making output if information is not present, coherence score (semantic similarity scoring: how well text align) and contextual relevance (response is focus on prompt) and perplexity to determine rate of generating/predicting novel word (abstractive summarization check)
- Determine the importance in RAG in text chatbots, improvement and limitation

2.1 Research Hypothesis/Questions

- The goal of the project is to create a LLM based text chatbot which incorporates RAG (Retrieval Augmented Generation). The project will investigate the role of RAG in hallucination based on the type of data (structured or unstructured data).
- Does RAG influence the rate of hallucination despite sufficient prompting?

2.2 Significance

- AI generates revenue for companies in all sectors by cutting time, saving cost and improving efficiency. AI role is not to replace the need for human interaction but to assist in optimizing decision making as data become more vast and too much to hand.
- According to CNBC, OpenAI expect to generate 12 billion by end of 2025 in total revenue, Anthropic generates about 3 billion in annualized revenue.

2.2.1 Organization of the Report

The rest of the report is structured as follows.

- Design and Methodology
- Proposed Method
- Tools and Technologies

- Data Gathering
- Data Analysis
- Results and Discussions
- Conclusion
- Future Scopes/Recommendations
- Related Works
- References

3 Design and Methodology

After inspecting the dataset. EDA was performed using TF-IDVectorizer to assess word similarity and key extraction via a statistical method (bag of words and TF-ID weighting) that balances frequency and and inverse frequency. Countvectorizer is used to count common words in the first 1500 words and displays the first 25 words. The result of the Countvectorizer is plotted on a bar chart while the result of the TF-IDvectorizer is visualized using a scatterplot from the matplotlib module/package

3.1 Proposed Method

The algorithm and AI framework used in this project was derived from openAI platform which hosted langchain, langchain-openai, faiss-cpu pypdf, bs4, and sentence-transformers. Other algorithms include libraries required for the evaluation metrics were rouge-score, pyTorch, sacrebleu, bert-score, torch transformers, ragas, deepeval

The flowchart below depicts the appropriate step and task that will be taken to complete the objective

- gpt-5-nano API model will be generated and the appropriate GPT model will be to train using large language model (LLM)
- Chunking will be done using recursive character text splitter with a chunk size of 250 and a chunk overlap of 10.
- For vector embedding the gpt-3.5-turbo, with a temperature of 0.8 and max token of 250 was used.
- Saves vectors into FAISS, allows you to find the most "similar" code instantly when a user asks a question.
- I Prompted with 4 different questions.
- For my demo, I used a machine learning python interface to visualize prompt and inspect text chat-bot generated output response

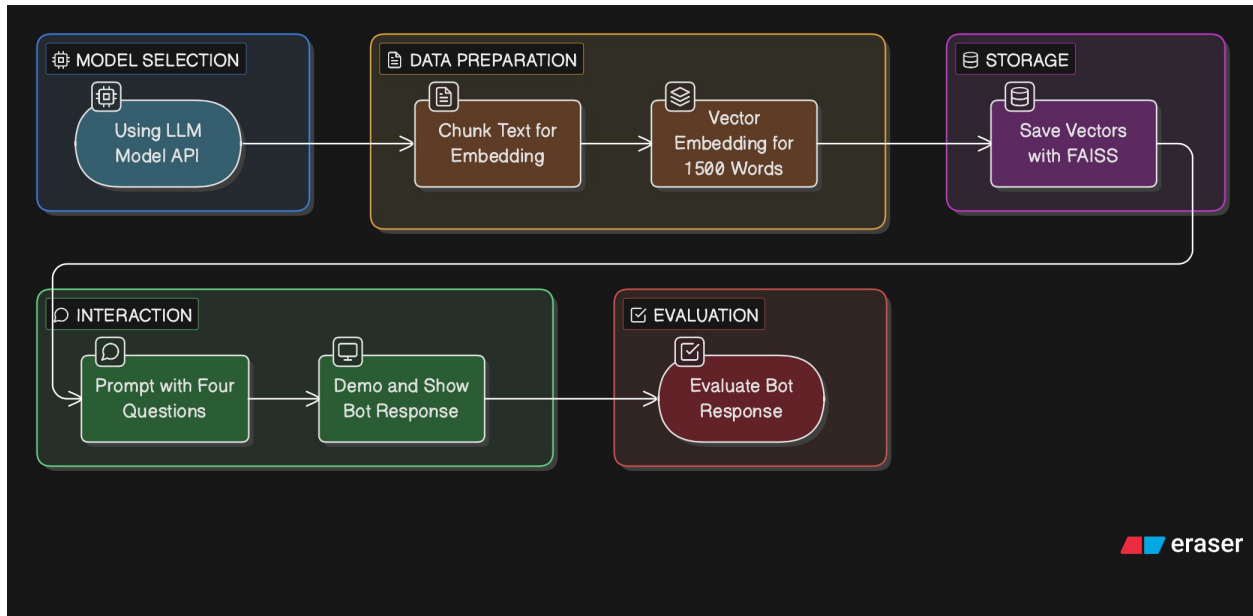


Figure 1: Flowchart showing the necessary events required to build a LLM based text chat-bot

- Aanalysis was done on the evaluated bot generated response to observe ground truth

3.2 Tools and Technologies

Some workflow tools used in completing this project include:

- OpenAI platform for generating API key and LLM based transformer model
- Python (Google Colab) for running the code and installing packages, modules and libraries
- Hugging Face for implementing Gradio and also adding an API key
- FAISS for storing vectors embedded (conversion of text to numerical)
- Gemini for insight, debugging, and information on RAG
- GitHub for the dataset
- Overleaf for documentation
- Microsoft Office for storing tabular data and creating a flowchart

3.3 Data Gathering

The dataset utilized in this project is a medical dataset derived from github, the dataset can also be found on Kaggle. The dataset consists of both a training and a test file of medical scripts by clinicians depicting symptoms, treatments and prognosis. The training data set is all in text format and is made up of 17,811,684 Characters and 2,612,370 words. For the inspection of RAG on my dataset, the first 4 paragraphs were trained using LLM which constitutes roughly 1500 words. The medical notes are divided into paragraphs based on specialty. The following specialty were

- Neoplasms
- Digestive system diseases
- Nervous system diseases
- Cardiovascular diseases
- General pathological conditions
- Github Link: <https://github.com/sebischair/medical-abstracts-tc-corpus?tab=readme-ov-file>

3.4 Data Analysis

Figure 1 provides details about the flowchart required to create an LLM based text-chat-bot. My analysis indicates that RAG is accurate in retrieving textual data but fails when prompting techniques are ambiguous. The observation from running various evaluation metrics is as seen below:

Figure 6 shows the ROUGE score based on recall using the F1-scores as a metrics which measure how much of the reference text is captured in the generated bot output. The scores observed were low (around 0.1, 0.2) and were excluded from data analysis due to very low accuracy

Figure 4 shows a BLEU score that indicates the precision of user response prompt based on how many n-grams in the machine output match the reference, the precision score was extremely low (almost zero across the board). A negative log graph was used to visualize the trend. Due to the extreme low accuracy, similar to the rouge score, it was excluded from analysis.

Figure 3 visualizes the BERT score for the response for all 4 user prompt measuring semantic closeness, this metrics showed a significant result which utilize both precision, recall and F1-score. Due to strong performance on this metrics, it will be used as the standard to evaluate my LLM based chat-bot prompt response.

Other metrics used were fallback rate which is trigger based on the fallback response that varies with how often you prompt the same question, perplexity score was 146.38 and the Coherence score was 0.8808

4 Results and Discussions

4.1 Discussion

BERT, ROUGE and BLEU scores measure precision and recall. BLEU centers on precision (which focuses on accuracy) while ROUGE centers on recall (exact matching) and is scored using the F1-metrics which is a balance of precision and recall to prevent verbose (cheating)

ROUGE score performed poorly based on how the metrics evaluate the bot generated output, ROUGE score matches exactly to the reference text that was used to train the bot (n-gram matching).

Similar to ROUGE score, BLEU score performed poorly because it is based on n-gram precision matching that matches context word for word. An LLM model has a temperature setting that is tied to word context.

A higher temperature will have similar context between the reference text and generated output despite not watching the exact n-gram which will always lead to a poor score on the ROUGE metrics.

There is a negative correlation between ROUGE and BLEU score with temperature, as temperature increase, these two metrics will always score low. Since my temperature

was set to 0.8. The results of both metrics were irrelevant and disregarded but added to show a comparison with BERT score

Focusing on BERT score each question prompt had different precision, recall and F1-Score. BERT score uses contextual embedding which is based on semantic similarity. The result derived is expected. Of the four user prompt, the second prompt scored high on all metrics as seen in the result question. Since each prompt is different, it is expected that the BERT score will change accordingly

A coherence score of the second question was found to be 0.897, the score indicates that the words within the bot generated response are highly related to one another semantically and is easy for human to understand.

A perplexity score of 146 was found using a gpt-2 large model. The perplexity score is higher than usual, which should hover around 20 -90. Perplexity score can be used as a measure of prompting techniques meaning if the user prompts a 15 word sentence and the training data span 1500 words, it could lead to confusion in the LLM model which is reflective of the score.

4.1.1 Limitations

Limitations pertains to the methodology, particularly chunking methods. A different chunking technique that was not explored could improve precision, recall and F1-Score on all evaluation metrics. Future research should include the ability to measure faithfulness to assess the truth for more complex chatbot that involves queries from agentic RAG. My hypothesis was to test a simple RAG setup by creating a LLM based chatbot. However, to properly assess the benefit of RAG in multimodal chatbots that receive complex queries from users, select tools, loop back to refine its own work and make decision that can affect human outcome. Faithfulness using Natural Language Inference (NLI) that measures hypothesis on entailment and contradiction is paramount (Gemini, 2025).

In a standard RAG setup, a single hallucination might just result in one bad answer. In an agentic workflow, it can trigger a catastrophic chain reaction.

4.1.2 Privacy

Some ethical, privacy and bias concerns that contains to the use of LLM based chatbots include the following

- The collection of data poses risk. Communication between a chatbot and user involves interaction between a user and the LLM.
- Chat logs could contain sensitive or personal queries such as feeling, mood or emotion. AI needs to analyze when to report or flag safety concerns

4.1.3 Risk of Hallucination can be detrimental

- Taking information as face value can lead to negative outcome, over-reliant of AI hallucinated as been linked to AI induced psychosis.

4.1.4 Ethics implementation

- Ethical rules will be incorporated in the training program to prevent the chatbot from making unfounded assumptions when prompted with queries requiring professional expertise.
- A fall back “If unsure, say “I don’t have enough information to answer that” or “I am not a licensed trained professional, but I can direct you to one is recommended.

5 Conclusions

The main takeaway of this project was to build a LLM based text chat-bot and investigate the use of RAG and LLM in retrieving and generating new response output. The project identified the retrieval component as one of the major source of hallucination during pilot testing of my chat-bot when generating output response. A relationship with poor user prompting techniques and generated output was also observed leading to a higher fall-back rate response. The hypothesis was supported showing a decrease in hallucination when using RAG native format which aligns well with unstructured text files.

A limitation and challenge encountered centered around effective chunking which emerged as one of the most important challenges in retrieving textualized unstructured data.

6 Future Scopes/Recommendations

My project can be easily reproduced for any text dataset, the dataset size need to be considered, appropriate chunking strategies must be incorporated based on size. As observed, hallucination is more prone to structured data in comparison to basic text data. Future research to improve the functionality of LLM based chatbot and minimize hallucination should focus on this issue. Some recommendations include:

- A focus on the textualization challenge involved in the conversion of structural data to textual format since RAG systems fundamentally operate on text. RAG needs to convert structured data into a textual format which introduce loss context and ambiguity for LLM
- A focus on dealing with schema Misinterpretation: Database require understanding relationships between tables or specific column. If the RAG system attempts to translate a natural language query into a database query (e.g., SQL), hallucination can occur.
- A solution to misassociation of values encountered by RAG: When similar values appear in different, unrelated contexts across various tables or rows, the textualized chunks might become very similar forcing RAG to incorrectly pull together these misassociated pieces of information, causing hallucination.

- Granularity Issues: Incorrect granularity (level of detail) can either provide too little context or too much noisy, irrelevant context, both of which can increase hallucination risk.

7 References

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8 Related Works

A Appendix

Hugging face: Gradio app could be installed on Colab temporarily but has to be hosted on hugging face. To do this, an API Key generated by OpenAI must be stored under has to be imported to hugging face, a new space is created which consists of a requirement text file that includes all necessary libraries and module stored under the main of the created space. The dataset must also be uploaded to main and the code should be migrated from Colab or other another IDE and stored under the app.py in the main section of the hugging face.

B Results

B.1 Figures and Tables Appendix

Table 2: BERT Score for all four questions prompted on the chatbot.

Model Name	Metrics	First Score	Second Score	Third Score	Fourth Score
Roberta-large	Precision	0.8677	0.874	0.804	0.8488
	F1	0.7799	0.7827	0.7679	0.7523
	Recall	0.8214	0.8259	0.7855	0.7976

Table 3: BLEU Score for all four questions prompted on the chatbot.

Model Name	Evaluation Metrics	First Score	Second Score	Third Score	Fourth Score
BLEU	Score	0.00322	0.02596	0.00452	1.02774

Table 4: Rouge score for the first 4 prompted questions outlining the precision, recall and F1 for three different models

Model Name	Metrics	Q1 Score	Q2 Score	Q3 Score	Q4 Score
Rouge 1	Precision	0.6538	0.5714	0.2647	0.3333
	Recall	0.0833	0.0784	0.0441	0.0147
	F1	0.1478	0.1379	0.0756	0.0282
Rouge 2	Precision	0.1600	0.4074	0.3016	0.9722
	Recall	0.3327	0.0247	0	0
	F1	0.0351	0.0542	0	0
Rouge 3	Precision	0.5000	0.5714	0.2059	0.3333
	Recall	0.0637	0.0784	0.0343	0.0147
	F1	0.1130	0.1379	0.0588	0.0282

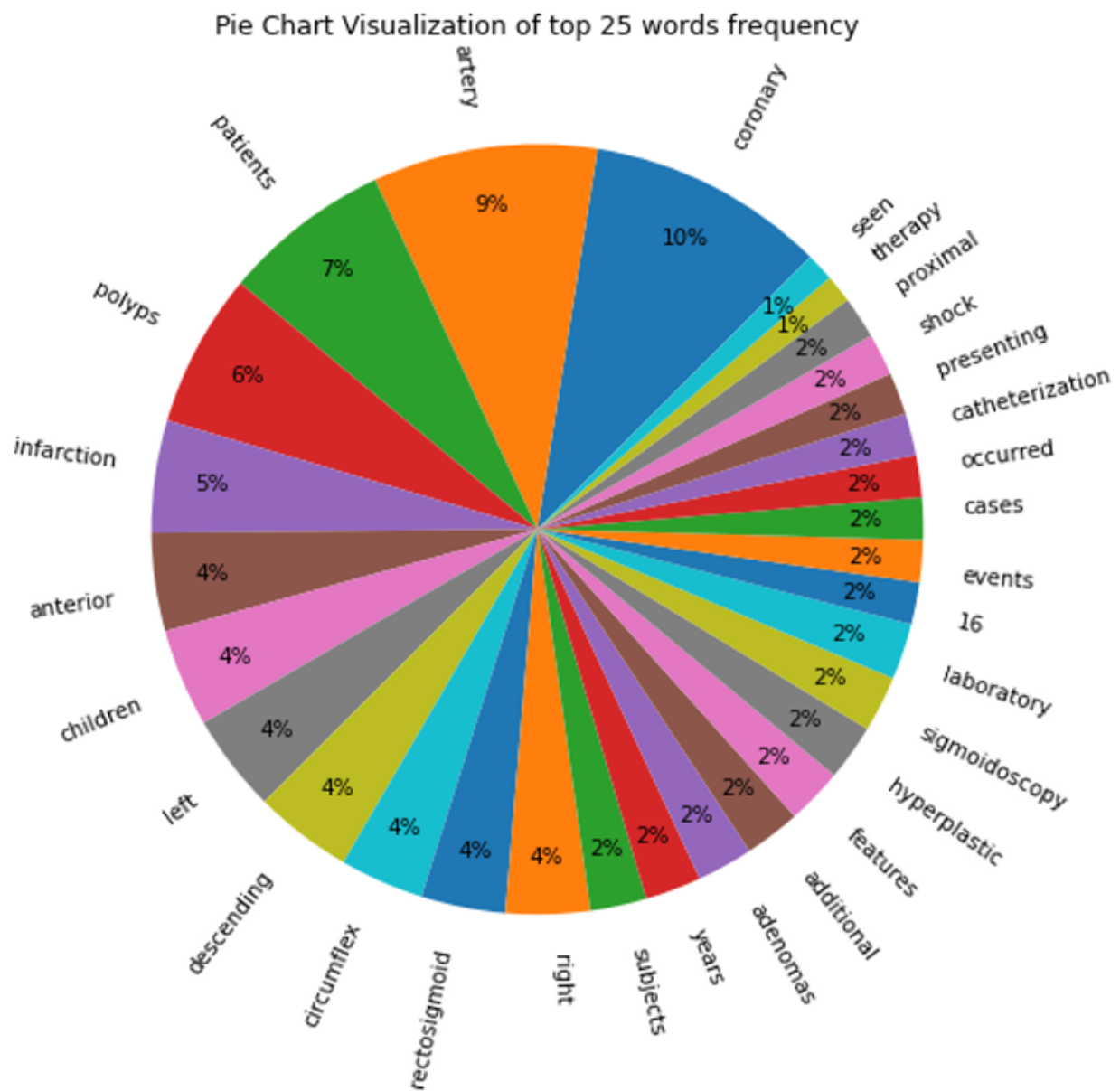


Figure 2: Pie Chart visualization of the top 25 words frequency and showing the percentage of this words

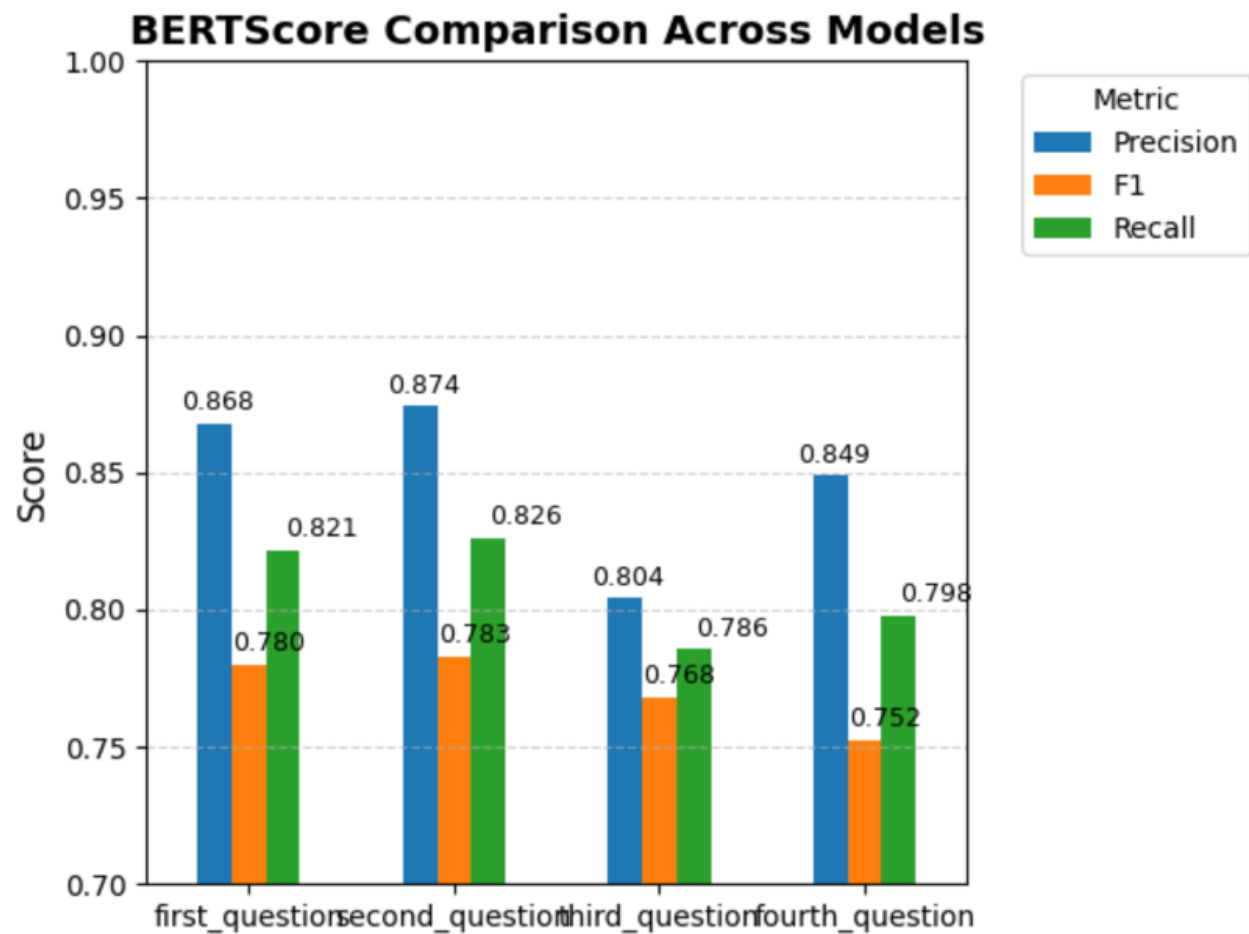


Figure 3: BERT Score comparison among the first three question to check for semantics closeness

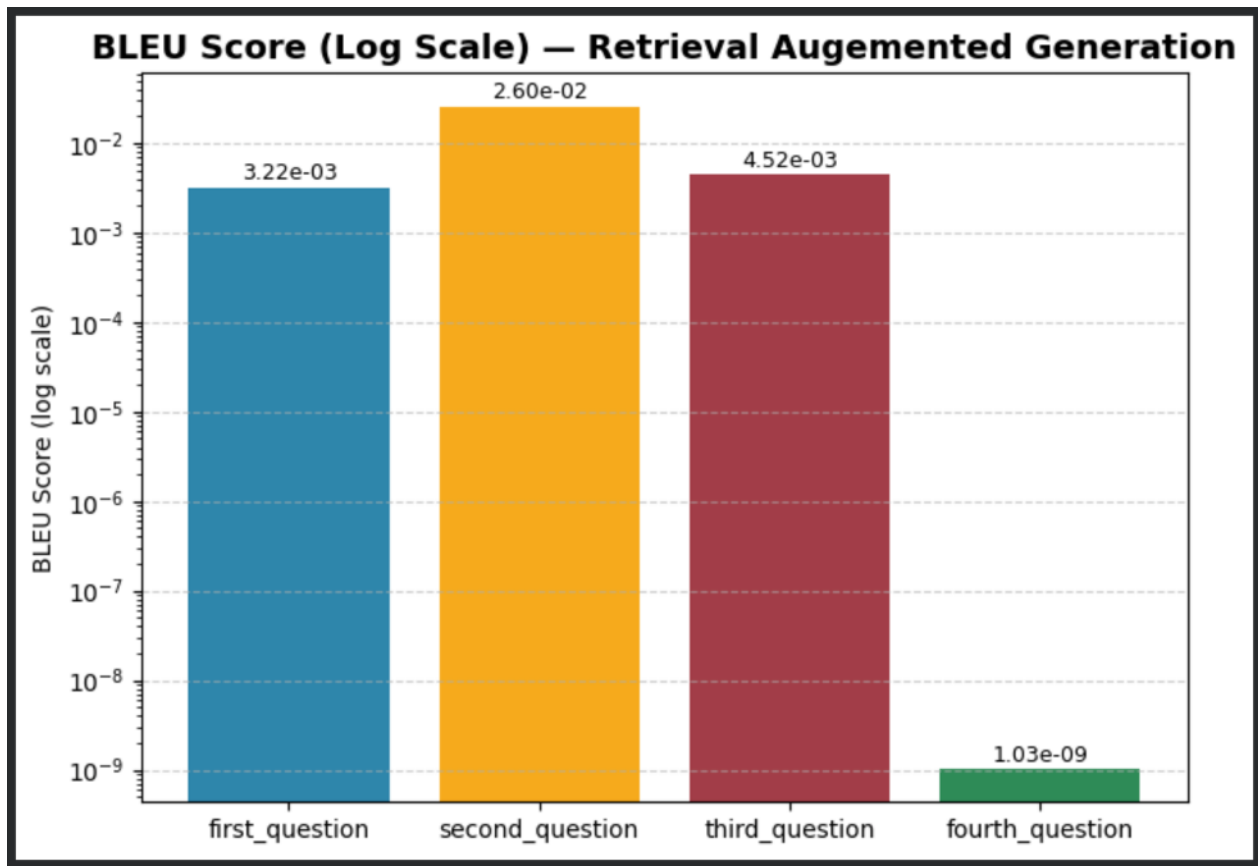


Figure 4: BLEU Score of all four questions plotted using an to determine the recall of chatbot response based on human response

```
%%time
first_question = ask_question("What were the ages of the patient in the Catheterization laboratory events and hospital outcome with direct angiople
print(first_question)

...
- The study included patients with acute myocardial infarction
- Patients who underwent direct infarct angioplasty were assessed for safety
- The study did not specify the exact ages of the patients in the results provided
CPU times: user 141 ms, sys: 3.17 ms, total: 144 ms
Wall time: 2.17 s

%%time
second_question = ask_question("What was the reason for conducting the Catheterization laboratory events?")
print(second_question )

The reason for conducting the Catheterization laboratory events was to assess the safety of direct infarct angioplasty without antecedent thrombolyt
CPU times: user 103 ms, sys: 1.2 ms, total: 104 ms
Wall time: 1 s

%%time
third_question = ask_question("What was the culprit of infection in renal abscesses in children for both male and female. How many cases were analyze
print(third_question )

* Staphylococcus aureus infections were identified in boys
* Escherichia coli infections were identified in girls
* A total of 26 cases were analyzed for the culprit of infection in renal abscesses in children
CPU times: user 101 ms, sys: 760 µs, total: 101 ms
Wall time: 1.06 s

%%time
fourth_question = ask_question("Polyps were found in how many subjects in the hyperplastic polyps?")
print(fourth_question)

* Polyps were found in 108 subjects with hyperplastic polyps.
CPU times: user 103 ms, sys: 891 µs, total: 104 ms
```

Figure 5: Screenshots of the 4 prompted question on my chatbot

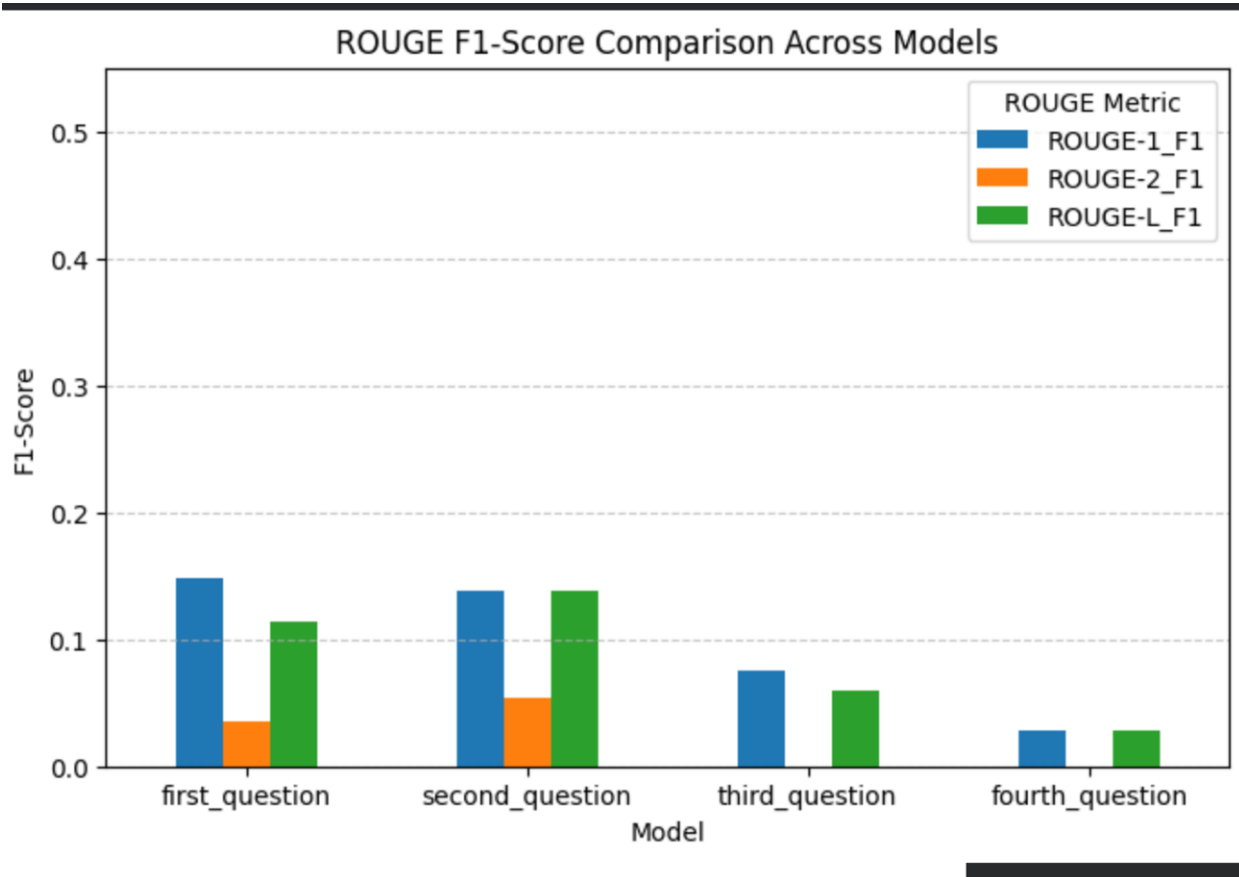


Figure 6: ROUGE Score of all four questions plotted using an to determine the recall of chatbot response based on human response

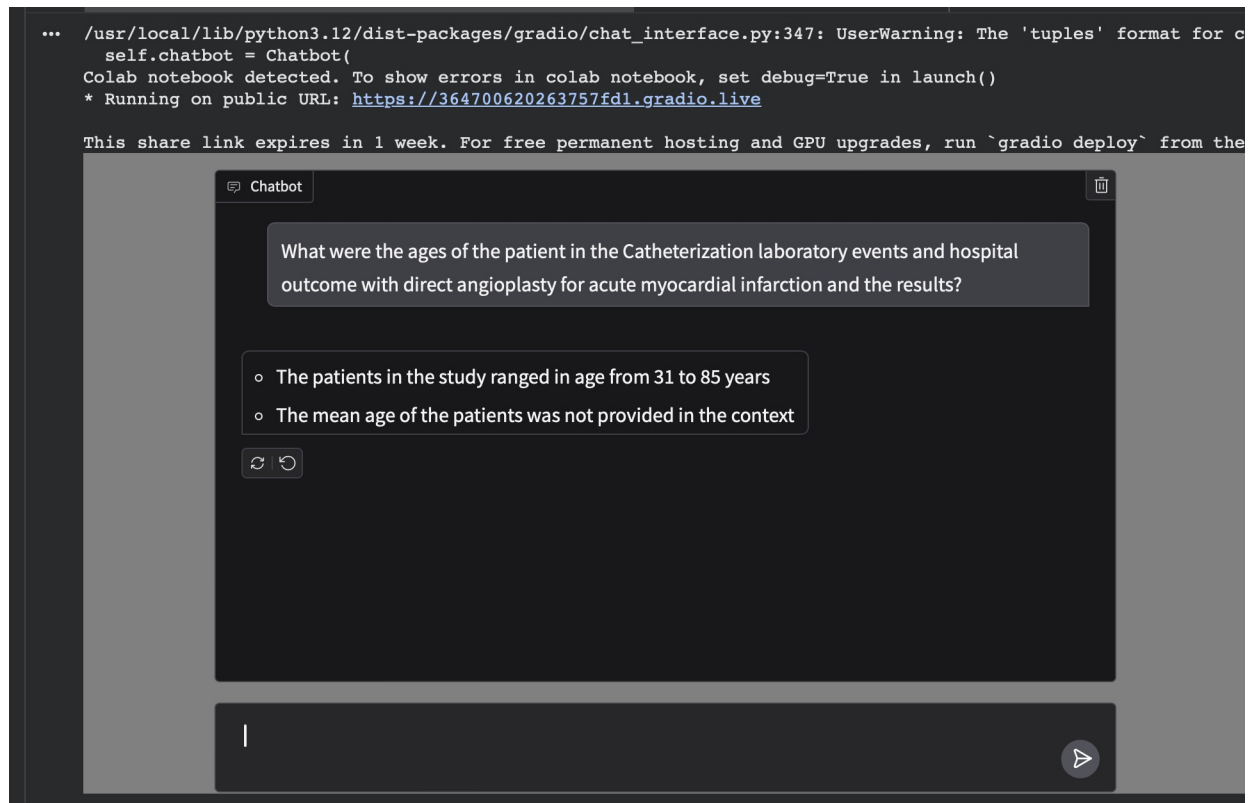


Figure 7: User prompt for question 1 on Gradio Chatbot

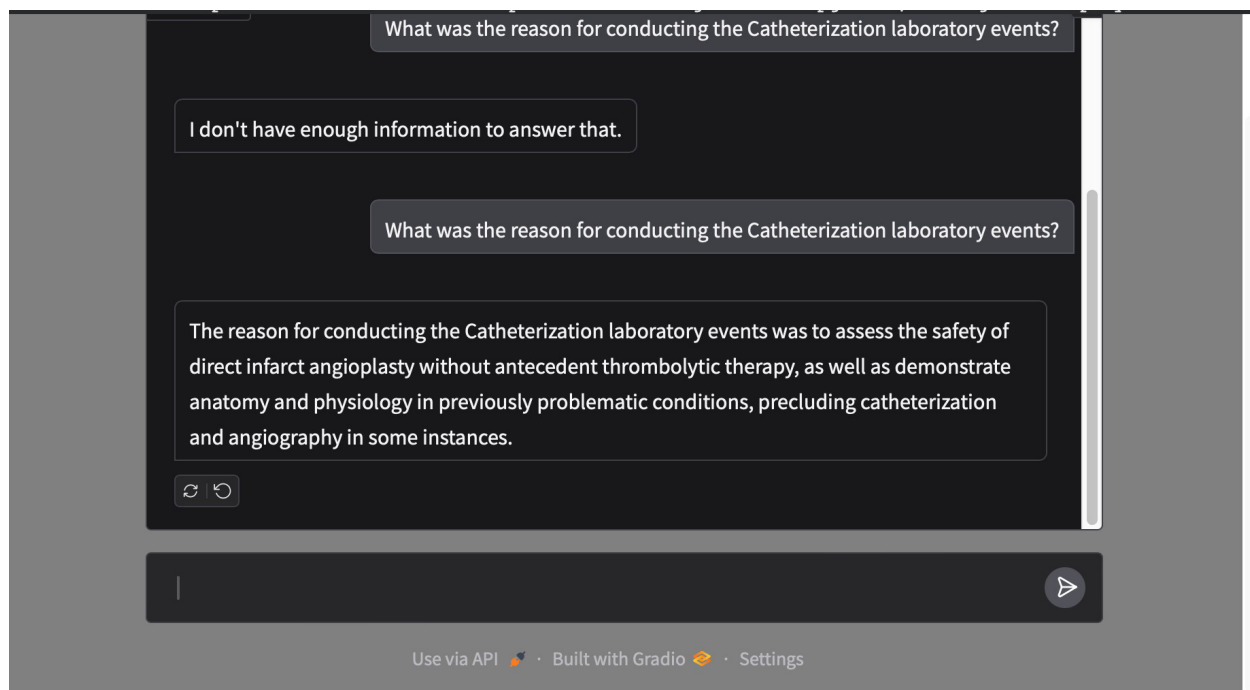


Figure 8: User prompt for question 2 on Gradio Chatbot

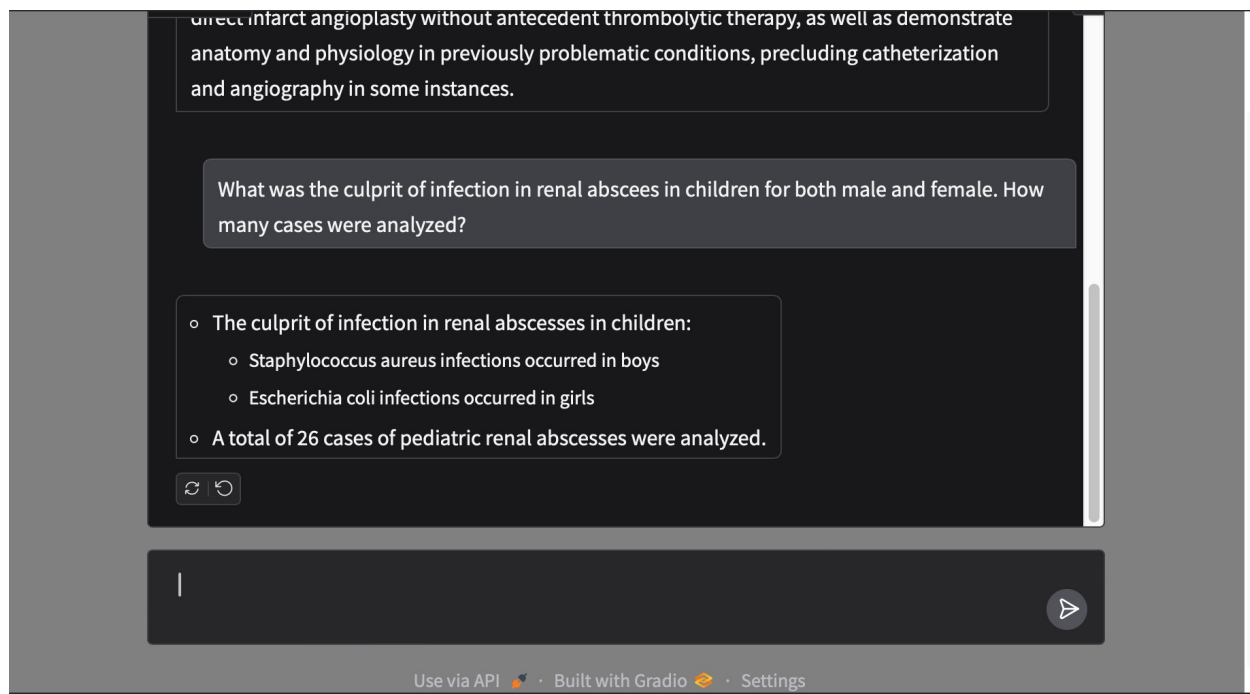


Figure 9: User prompt for question 3 on Gradio Chatbot

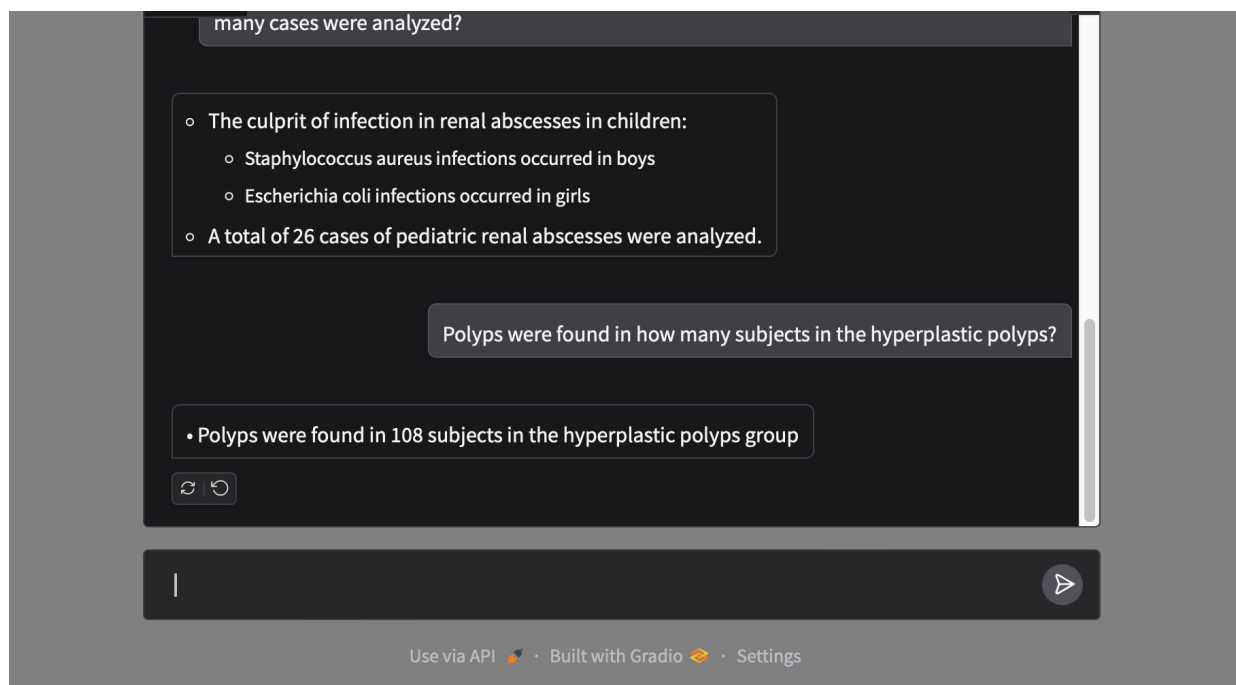


Figure 10: User prompt for question 4 on Gradio Chatbot

TFID Vectorizer showing the importance of the top 30 word to the first 3 paragraph while penalizing common words in determining semantic structure

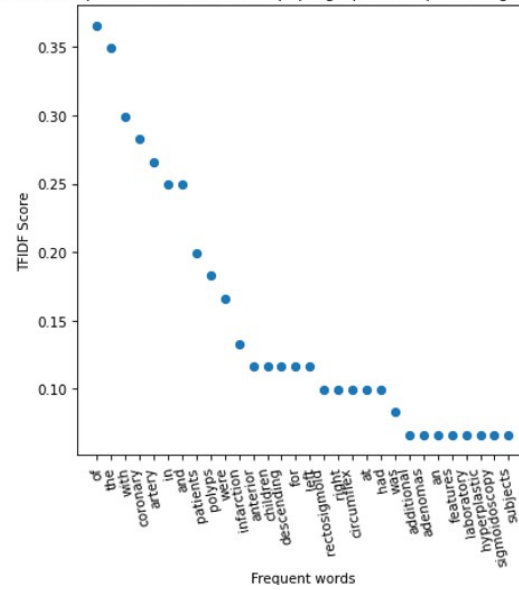


Figure 11: TF-IDVectorizer showing the importance of the top 30 words while penalizing common words